

```
> l[["name"]]
[1] "R"
```

A list can be converted to a vector using the `unlist()` function. The assignment operator can be used to update a value in a list. An example is given below:

```
> l <- list("R", 28, "GPLV2")
> l[3] <- "GNU GPLV2"
> l
[[1]]
[1] "R"

[[2]]
[1] 28

[[3]]
[1] "GNU GPLV2"
```

Factors

A factor is an ordered collection of entities that are used to represent categories. Although you can use vectors for the same, these can be inefficient for large data. You can use the `factor()` function to create a factor. For example:

```
> gender <- factor(c("male", "female", "male", "female",
"male"))
> gender
[1] male   female male   female male
Levels: female male
```

The different values that a factor can take are known as levels. The factors are actually stored internally as integer vectors. This can be seen using the `str()` function:

```
> str(gender)
Factor w/ 2 levels "female","male": 2 1 2 1 2
```

If you try to apply the `min()` function for the above categorical values, R will throw an error. You need to use the ordered flag to indicate that there is a logical ordering to the categories. For example:

```
> room <- factor(c("standard", "premium", "luxury"), ordered
= TRUE)
> room
[1] standard premium luxury
Levels: luxury < premium < standard
```

The various levels for the room example are shown below:

```
> levels(room)
[1] "luxury" "premium" "standard"

> nlevels(room)
[1] 3
```

You will now be able to apply the `min()` function to the room factor to return valid levels, as illustrated below:

```
> min(room)
[1] luxury
Levels: luxury < premium < standard
```

Data frames

A data frame contains a number of named vectors that have the same length. These frames are internally built on R lists. The `mtcars` data set in R provides Motor Trend Car Road Test information from the 1974 *Motor Trend* US magazine (Figure 1). It has information on 11 numeric values such as miles/ (US) gallon (mpg), number of cylinders (cyl), displacement (cu.in) (disp), horsepower (hp), rear axle ratio (drat), weight (1000 lbs) (wt), 1/4 mile time (qsec), engine (0 = V-shaped, 1 = straight) (vs), transmission (0 = automatic, 1 = manual) (am), number of gears (gear) and carburetor (carb).

```
> mtcars
      mpg  cyl  disp  hp drat   wt  qsec vs am gear carb
Mazda RX4    21.0   6 160.0 110 3.90 2.620 16.46 0   1   4   4
Mazda RX4 Wag 21.0   6 160.0 110 3.90 2.875 17.02 0   1   4   4
Datsun 710    22.8   4 108.0  93 3.85 2.320 18.61 1   1   4   1
Hornet 4 Drive 21.4   6 258.0 110 3.08 3.215 19.44 1   0   3   1
Hornet Sportabout 18.7   8 360.0 175 3.15 3.440 17.02 0   0   3   2
Valiant      18.1   6 225.0 105 2.76 3.460 20.22 1   0   3   1
Duster 360   14.3   8 360.0 245 3.21 3.570 15.84 0   0   3   4
Merc 240D     24.4   4 146.7  62 3.69 3.190 20.00 1   0   4   2
Merc 230      22.8   4 140.8  95 3.92 3.150 22.90 1   0   4   2
Merc 280      19.2   6 167.6 123 3.92 3.440 18.30 1   0   4   4
Merc 280C     17.8   6 167.6 123 3.92 3.440 18.90 1   0   4   4
Merc 450SE    16.4   8 275.8 180 3.07 4.070 17.40 0   0   3   3
Merc 450SL    17.3   8 275.8 180 3.07 3.730 17.60 0   0   3   3
Merc 450SLC   15.2   8 275.8 180 3.07 3.780 18.00 0   0   3   3
Cadillac Fleetwood 10.4   8 472.0 205 2.93 5.250 17.98 0   0   3   4
Lincoln Continental 10.4   8 460.0 215 3.00 5.424 17.82 0   0   3   4
Chrysler Imperial 14.7   8 440.0 230 3.23 5.345 17.42 0   0   3   4
Fiat 128      32.4   4  78.7  66 4.08 2.200 19.47 1   1   4   1
Honda Civic   30.4   4  75.7  52 4.93 1.615 18.52 1   1   4   2
Toyota Corolla 33.9   4  71.1  65 4.22 1.835 19.90 1   1   4   1
Toyota Corona 21.5   4 120.1  97 3.70 2.465 20.01 1   0   3   1
Dodge Challenger 15.5   8 318.0 150 2.76 3.520 16.87 0   0   3   2
AMC Javelin   15.2   8 304.0 150 3.15 3.435 17.30 0   0   3   2
Camaro Z28    13.3   8 350.0 245 3.73 3.840 15.41 0   0   3   4
Pontiac Firebird 19.2   8 400.0 175 3.08 3.845 17.05 0   0   3   2
Fiat X1-9     27.3   4  79.0  66 4.08 1.935 18.90 1   1   4   1
Porsche 914-2 26.0   4 120.3  91 4.43 2.140 16.70 0   1   5   2
Lotus Europa  30.4   4  95.1 113 3.77 1.513 16.90 1   1   5   2
Ford Pantera L 15.8   8 351.0 264 4.22 3.170 14.50 0   1   5   4
Ferrari Dino  19.7   6 145.0 175 3.62 2.770 15.50 0   1   5   6
Maserati Bora  15.0   8 301.0 335 3.54 3.570 14.60 0   1   5   8
Volvo 142E    21.4   4 121.0 109 4.11 2.780 18.60 1   1   4   2
```

Figure 1: The `mtcars` data set

We can create a data frame with a sample set from the `mtcars` data set as follows:

```
> name <- c("Mazda RX4", "Mazda RX4 Wag", "Datsun 710",
"Hornet 4 Drive", "Hornet Sportabout", "Valiant")
> name
```